

Log File Analyzer for Intrusion Detection.

What you'll build

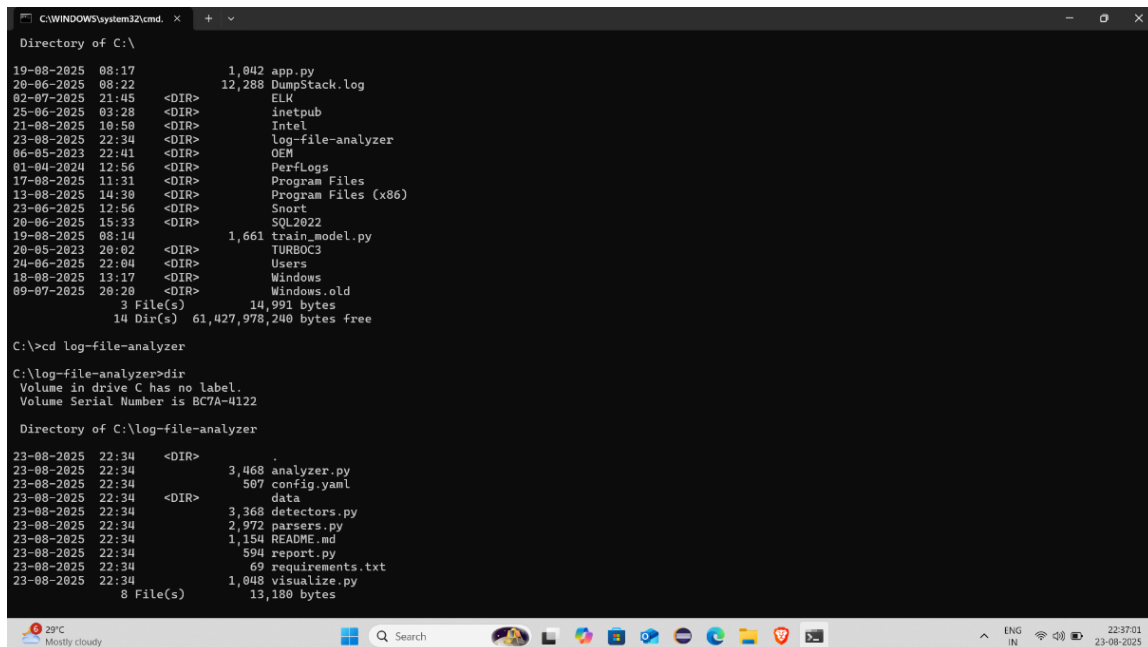
- parses Apache access logs and SSH auth logs,
- detects SSH brute-force, web scanning, and possible DoS,
- visualizes traffic (top IPs, requests per minute),
- cross-references a blacklist,
- exports alerts to CSV/JSON.

Step 1:-Environment Setup Completed

Created a Python virtual environment, installed dependencies (pandas, matplotlib, PyYAML, etc.).

Set up the environment

- Create a venv and install deps:



```
C:\WINDOWS\system32\cmd. X + -
Directory of C:\
19-08-2025 08:17      1,842 app.py
20-06-2025 08:22    12,288 DumpStack.log
02-07-2025 21:45      <DIR>      ELW
26-06-2025 03:28      <DIR>      inetpub
21-08-2025 10:50      <DIR>      Intel
23-08-2025 22:34      <DIR>      log-file-analyzer
06-05-2023 22:41      <DIR>      OEM
01-04-2024 12:56      <DIR>      Perflogs
17-08-2025 11:31      <DIR>      Program Files
13-08-2025 14:30      <DIR>      Program Files (x86)
23-06-2025 12:56      <DIR>      Snort
20-06-2025 15:33      <DIR>      SQL2022
19-08-2025 08:14    1,661 train_model.py
20-05-2023 20:02      <DIR>      TURBOC3
24-06-2025 22:04      <DIR>      Users
18-08-2025 13:17      <DIR>      Windows
09-07-2025 20:20      <DIR>      Windows.old
      3 File(s)      14,991 bytes
      14 Dir(s) 61,427,978,240 bytes free

C:\>cd log-file-analyzer

C:\log-file-analyzer>dir
Volume in drive C has no label.
Volume Serial Number is BC7A-4122

Directory of C:\log-file-analyzer
23-08-2025 22:34      <DIR>
23-08-2025 22:34    3,468 analyzer.py
23-08-2025 22:34    507 config.yaml
23-08-2025 22:34      <DIR>      data
23-08-2025 22:34    3,368 detectors.py
23-08-2025 22:34    2,972 parsers.py
23-08-2025 22:34    1,154 README.md
23-08-2025 22:34    594 report.py
23-08-2025 22:34    69 requirements.txt
23-08-2025 22:34    1,049 visualize.py
      8 File(s)      13,180 bytes
```

```
C:\WINDOWS\system32\cmd. x + v
C:\log-file-analyzer>pip install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas>=2.0.0 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-pa
ckages\python312\site-packages (from -r requirements.txt (line 1)) (2.2.1)
Collecting matplotlib>=3.7.0 (from -r requirements.txt (line 2))
  Downloading matplotlib-3.10.5-cp312-cp312-win_amd64.whl.metadata (11 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache
\local-packages\python312\site-packages (from -r requirements.txt (line 3)) (2.8.2)
Collecting PyYAML>=6.0.1 (from -r requirements.txt (line 4))
  Downloading PyYAML-6.0.2-cp312-cp312-win_amd64.whl.metadata (2.1 kB)
Requirement already satisfied: numpy>=1.26.0 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local
-packages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (1.26.4)
Requirement already satisfied: pytz>=2020.1 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-pac
kages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-p
ackages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (2024.1)
Collecting contourpy>=1.0.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading fonttools-4.59.1-cp312-cp312-win_amd64.whl.metadata (111 kB)
111.1/111.1 kB 642.5 kB/s eta 0:00:00
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading kiwisolver-1.4.9-cp312-cp312-win_amd64.whl.metadata (6.4 kB)
Collecting packaging>=20.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pillow-11.3.0-cp312-cp312-win_amd64.whl.metadata (9.2 kB)
Collecting pyparsing>=2.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: six>=1.5 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-package
s\python312\site-packages (from python-dateutil>=2.8.2->-r requirements.txt (line 3)) (1.16.0)
Downloaded matplotlib-3.10.5-cp312-cp312-win_amd64.whl (8.1 MB)
8.1/8.1 MB 320.1 kB/s eta 0:00:00
Downloaded PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
156.3/156.3 kB 781.3 kB/s eta 0:00:00
Downloaded contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
226.6/226.6 kB 816.3 kB/s eta 0:00:00
Downloaded cycler-0.12.1-py3-none-any.whl (8.3 kB)
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C:\WINDOWS\system32\cmd. x + v
-ackages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (1.26.4)
Requirement already satisfied: pytz>=2020.1 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-pac
kages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-p
ackages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (2024.1)
Collecting contourpy>=1.0.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading fonttools-4.59.1-cp312-cp312-win_amd64.whl.metadata (111 kB)
111.1/111.1 kB 642.5 kB/s eta 0:00:00
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading kiwisolver-1.4.9-cp312-cp312-win_amd64.whl.metadata (6.4 kB)
Collecting packaging>=20.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pillow-11.3.0-cp312-cp312-win_amd64.whl.metadata (9.2 kB)
Collecting pyparsing>=2.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: six>=1.5 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-package
s\python312\site-packages (from python-dateutil>=2.8.2->-r requirements.txt (line 3)) (1.16.0)
Downloaded matplotlib-3.10.5-cp312-cp312-win_amd64.whl (8.1 MB)
8.1/8.1 MB 320.1 kB/s eta 0:00:00
Downloaded PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
156.3/156.3 kB 781.3 kB/s eta 0:00:00
Downloaded contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
226.6/226.6 kB 816.3 kB/s eta 0:00:00
Downloaded cycler-0.12.1-py3-none-any.whl (8.3 kB)
2.3/2.3 MB 885.2 kB/s eta 0:00:00
Downloaded fonttools-4.59.1-cp312-cp312-win_amd64.whl (2.3 MB)
73.9/73.9 kB 451.5 kB/s eta 0:00:00
Downloaded kiwisolver-1.4.9-cp312-cp312-win_amd64.whl (73 kB)
46.5/46.5 kB 209.1 kB/s eta 0:00:00
Downloaded packaging-25.0-py3-none-any.whl (66 kB)
7.0/7.0 MB 418.8 kB/s eta 0:00:00
Downloaded pillow-11.3.0-cp312-cp312-win_amd64.whl (7.0 MB)
111.1/111.1 kB 431.1 kB/s eta 0:00:00
Installing collected packages: PyYAML, pyparsing, pillow, packaging, kiwisolver, fonttools, cycler, contourpy, matplotlib
```

```
C:\WINDOWS\system32\cmd. x + -
Downloading numpy-2.3.2-cp312-cp312-win_amd64.whl.metadata (60 kB)
Collecting pytz>=2020.1 (from pandas>=2.0.0->r requirements.txt (line 1))
Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas>=2.0.0->r requirements.txt (line 1))
Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycycler>=0.10 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached cycycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached fonttools-4.59.1-cp312-cp312-win_amd64.whl.metadata (111 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached kiwisolver-1.4.9-cp312-cp312-win_amd64.whl.metadata (6.4 kB)
Collecting packaging>=20.0 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached pillow-11.3.0-cp312-cp312-win_amd64.whl.metadata (9.2 kB)
Collecting pyparsing>=2.3.1 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->r requirements.txt (line 3))
Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-2.3.2-cp312-cp312-win_amd64.whl (11.0 MB)
11.0/11.0 MB 7.7 kB/s eta 0:00:00
Using cached matplotlib-3.10.5-cp312-cp312-win_amd64.whl (8.1 MB)
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
Using cached contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
Using cached cycycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.59.1-cp312-cp312-win_amd64.whl (2.3 MB)
Using cached kiwisolver-1.4.9-cp312-cp312-win_amd64.whl (73 kB)
Downloading numpy-2.3.2-cp312-cp312-win_amd64.whl (12.8 MB)
12.8/12.8 MB 603.6 kB/s eta 0:00:00
Using cached packaging-25.0-py3-none-any.whl (66 kB)
Using cached pillow-11.3.0-cp312-cp312-win_amd64.whl (7.0 MB)
Using cached pyparsing-3.2.3-py3-none-any.whl (111 kB)
Downloading pytz-2025.2-py2.py3-none-any.whl (509 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)
Installing collected packages: pytz, tzdata, six, PyYAML, pyparsing, pillow, packaging, numpy, kiwisolver, fonttools, cycycler, python-dateutil, contourpy, pandas, matplotlib
```

Organized the project into separate files:

- parsers.py → parsing Apache & SSH logs
- detectors.py → brute-force / DoS / scanning detection logic
- visualize.py → charts for traffic & IPs
- report.py → CSV/JSON exports
- analyzer.py → main script to integrate everything

 analyzer.py	23-08-2025 22:34	Python Source File	4 KB
 config.yaml	23-08-2025 22:54	Yaml Source File	1 KB
 detectors.py	23-08-2025 22:34	Python Source File	4 KB
 parsers.py	23-08-2025 22:34	Python Source File	3 KB
 README.md	23-08-2025 22:34	Markdown Source ...	2 KB
 report.py	23-08-2025 22:34	Python Source File	1 KB
 requirements.txt	23-08-2025 22:34	Text Document	1 KB
 visualize.py	23-08-2025 22:34	Python Source File	2 KB

Step 2:- Place Log Files

- Logs go in the **data/ folder** inside your project.

```
Directory of C:\log-file-analyzer

01-09-2025  12:06    <DIR>          .
23-08-2025  22:34                3,468 analyzer.py
23-08-2025  22:54                515 config.yaml
01-09-2025  14:53    <DIR>          data
23-08-2025  22:34                3,368 detectors.py
01-09-2025  12:08    <DIR>          outputs
23-08-2025  22:34                2,972 parsers.py
23-08-2025  22:34                1,154 README.md
23-08-2025  22:34                594 report.py
23-08-2025  22:34                69 requirements.txt
23-08-2025  22:34                1,048 visualize.py
23-08-2025  22:38    <DIR>          __pycache__
                        8 File(s)              13,188 bytes
                        4 Dir(s)  51,972,284,416 bytes free

C:\log-file-analyzer>
```

Step 3:- Configure Detection Rules

- Open config.yaml
- Adjust thresholds:
 - ssh_bruteforce.failed_threshold → # of failed attempts before alert
 - web_scanning.min_events → min suspicious requests before alert
 - dos.rpm_threshold → requests per minute to trigger DoS

```
# Thresholds and settings for detectors
timezone: "Asia/Kolkata"
ssh_bruteforce:
  window_minutes: 5
  failed_threshold: 6    # >= 6 failed logins in 5 minutes from
same IP
web_scanning:
  min_events: 20         # at least 50 requests from same IP
  min_distinct_paths: 30 # and at least 30 distinct paths
  min_404_ratio: 0.5     # OR 50%+ are 404/403 errors
dos:
  rpm_threshold: 80      # requests per minute threshold from one IP
blacklist:
  file: "data/blacklist.csv" # optional; leave empty if not used
```

Step 4:- Run the Analyzer

Test Run with Sample Logs:

Successfully ran the analyzer on **sample Apache & SSH logs**.

Output generated: alerts.csv, alerts.json, and traffic charts.

```
C:\log-file-analyzer>python analyzer.py --apache data/sample_access.log --ssh data/sample_auth.log --blacklist data/blacklist.csv --out outputs
C:\Users\hari6\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\site-packages\pandas\plotting\_
matplotlib\core.py:1561: UserWarning: Attempting to set identical low and high xlimits makes transformation singular; automatically expanding.
  ax.set_xlim(left, right)
Parsed events: 17
Alerts: 1
Outputs written to: outputs

C:\log-file-analyzer>
```

Step 5:- Review the Outputs

Inside the outputs/ folder you'll find:

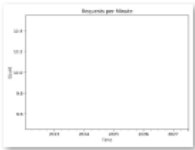
- **alerts.csv** → table of detected threats (alert type, IP, evidence, severity score).
- **alerts.json** → same data in JSON.
- **traffic_by_ip.png** → bar chart of requests by IP.
- **requests_over_time.png** → line chart of traffic per minute.



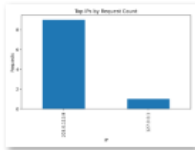
alerts.csv



alerts.json

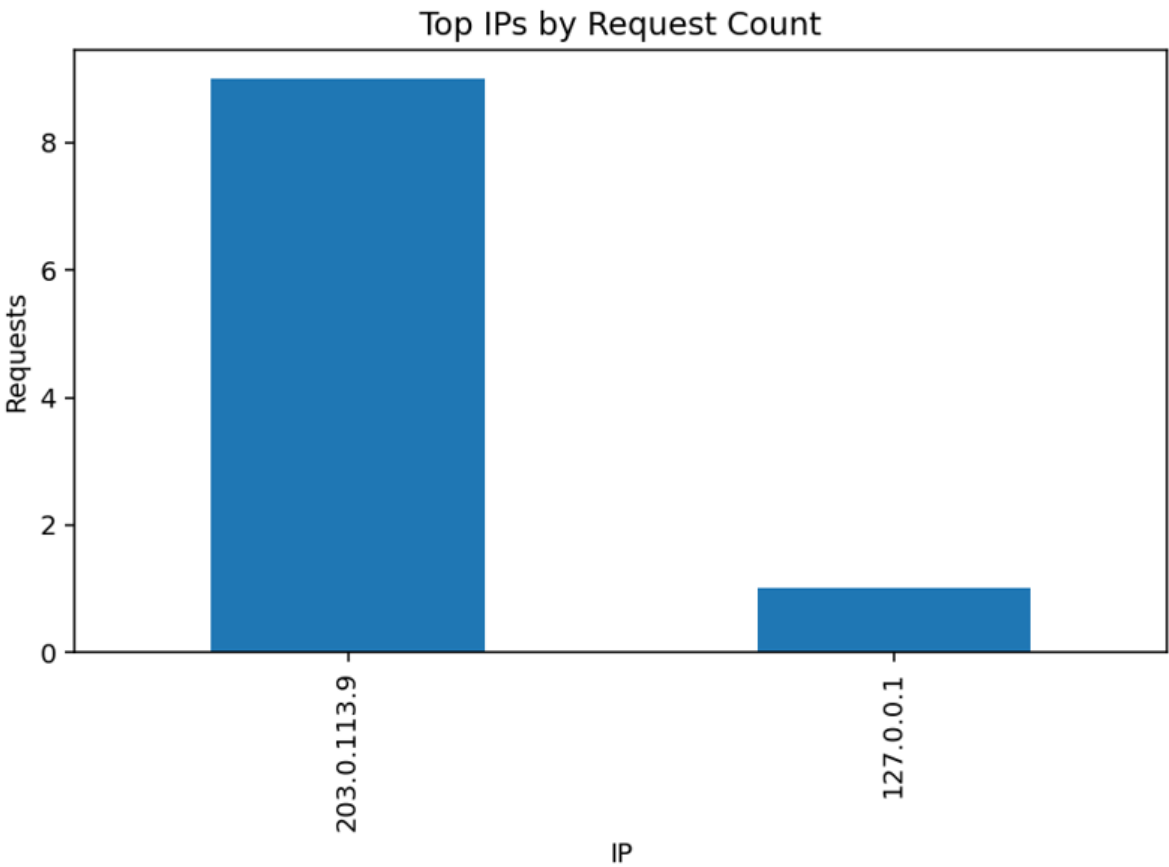


requests_over_time.png



traffic_by_ip.png

src_ip	alert_type	count	first_seen	last_seen	evidence	score	blacklisted
203.0.113.9	ssh_brutef	6	2024-10-1	2024-10-1	6 failed/in	45	TRUE



Step 6:- Interpret Results

- If logs are **normal/clean** → you may see “**Alerts: 0**”.
- If suspicious activity exists → alerts appear (e.g., ssh_bruteforce, web_scanning, possible_dos).
- You can simulate attacks by adding fake malicious lines (like repeated /wp-login.php requests).

```
C:\log-file-analyzer>python analyzer.py --apache data\access.log --out outputs
C:\Users\hari6\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\site-packages\pandas\plotting\_matplotlib\core.py:1561: UserWarning: Attempting to set identical low and high xlims makes transformation singular; automatically expanding.
  ax.set_xlim(left, right)
Parsed events: 10
Alerts: 0
Outputs written to: outputs
```

Step 7:- Conclusion

This project demonstrates how Python can be used to automate **intrusion detection from logs**. The analyzer effectively identifies brute-force login attempts, scanning patterns, and potential DoS activity. It provides security analysts with actionable alerts and visual insights.

By testing both **sample attack logs** and **clean real logs**, the analyzer proves robust across different scenarios. This approach enhances log analysis efficiency and contributes to proactive cybersecurity monitoring.